

# Probability

A review

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**What is Probability?** Probability measures **how likely** something is to happen, from:

- **0** (never happens)
- to **1** (always happens) (or 0% to 100%)

Rolling a fair die:

- $P(\text{Even}) = \frac{3}{6} = 0.5$
- $P(\text{Prime}) = \frac{3}{6} = 0.5$

## Basic Probability Formula

$$P(\text{Event}) = \frac{\text{Number of ways it can happen}}{\text{Total possible outcomes}}$$

## Important rules

### Certain Event (100%)

$$P(\text{Sun rises}) = 1$$

### Impossible Event (0%)

$$P(\text{Rolling a 7 on standard die}) = 0$$

### Complement Rule ("Not")

$$P(\text{Not } A) = 1 - P(A)$$

### Conditional Probability ("If...Then")

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$$

## Rules of probability

| Rule               | Formula                                   |
|--------------------|-------------------------------------------|
| Complement         | $P(A^c) = 1 - P(A)$                       |
| Union              | $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ |
| Mutually Exclusive | $P(A \cap B) = 0$                         |
| Independent        | $P(A \cap B) = P(A)P(B)$                  |

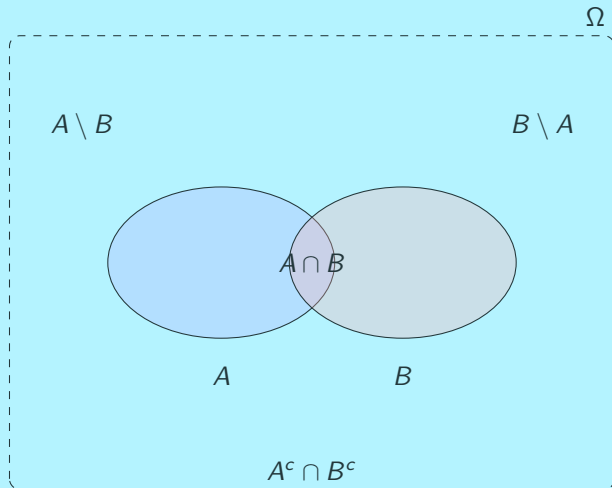
## Bayes' Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

## Bayes' Theorem originates from Conditional Probability

The probability of  $A$  *given*  $B$ :

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$



## Real-world uses

- Medical diagnosis
- Spam filtering
- Risk assessment
- Machine learning (Naive Bayes)

## Key Insight

- Conditional probability formalizes how we should rationally update beliefs given evidence.



## Example 1

A family has two children. You know that at least one of them is a boy. What is the probability that both children are boys?

## Example 1 - solution

**Step 1: Sample Space** The possible gender combinations for two children (older child first) are:

$$S = \{BB, BG, GB, GG\}$$

**Step 2: Apply Condition** We know *at least one* is a boy, so we eliminate GG:

$$S' = \{BB, BG, GB\}$$

**Step 3: Count Favorable Outcomes** Only BB has two boys:

$$\text{Favorable} = 1 \quad (BB)$$

$$\text{Possible} = 3 \quad (BB, BG, GB)$$

**Step 4: Calculate Probability**

$$P(\text{Both boys}) = \frac{\text{Favorable}}{\text{Possible}} = \frac{1}{3}$$

## Example 2

A bag contains 4 red marbles, 5 blue marbles, and 6 green marbles. If 3 marbles are randomly selected without replacement, what is the probability that all three are blue?

- $\frac{1}{91}$
- $\frac{2}{91}$
- $\frac{5}{182}$
- $\frac{10}{273}$
- $\frac{25}{546}$

## Example 2 - solution

**Step 1: Determine Total Marbles** First, find the total number of marbles:

$$4 \text{ red} + 5 \text{ blue} + 6 \text{ green} = 15 \text{ marbles}$$

**Step 2: Calculate Total Possible Outcomes** Number of ways to choose any 3 marbles from 15:

$$\text{Total combinations} = \binom{15}{3} = \frac{15!}{3!(15-3)!} = 455$$

**Step 3: Calculate Favorable Outcomes** Number of ways to choose 3 blue marbles from 5 available:

$$\text{Favorable combinations} = \binom{5}{3} = \frac{5!}{3!(5-3)!} = 10$$

**Step 4: Compute Probability**

$$P(3 \text{ blue}) = \frac{\text{Favorable}}{\text{Total}} = \frac{10}{455} = \frac{2}{91}$$

**Next class we will dive into Expectations and Variances**